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An investigation of primate life-history database quality

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INTRODUCTION

- Life history theory provides us with a lexicon of variables that characterize the timing and durations of growth and reproduction. [1, 2]
- Quantifying life history traits requires **abundant, high-quality, longitudinal data** to observe and infer species-level rates. [1, 3, 4]
- Life history strategies [3], complicating investigations when **some traits are unrecorded** for a focal species.
- Researchers face consistent **methodological challenges** in collating LH data for inter-species comparisons [1, 3, 4, 5], such as:
 - Discrepancies in data quality & trait inclusion
 - Taxonomic representation
 - Differences in variable definitions
 - Choosing appropriate variables for their study

OBJECTIVES

- To examine the primate LH characteristics of the Ernest [6], AnAge [7, 8], PanTHERIA [9], Kappeler & Pereira [10], and Amniotes [11] databases.
- To address consistency and usage issues for researchers who employ life history data
- To offer recommendations for best practices when using collated primate LH data

METHODS

- We identified and selected all datasets that included wide life history trait coverage for primates [5].
- We compared the structures of the datasets and selected seven key life history characteristics.
- We compared these datasets using several metrics:
 - The presence of values to evaluate missingness
 - Taxonomic representation to measure bias in species coverage
 - Metadata accessibility and the detail and consistency of variable descriptions

Table 1. Descriptive statistics of primate life-history characteristics for each database. AnAge is the only database that is continually updated, with its last update in 2023. Each database has fewer missing values for our selected key life-history traits compared to all recorded statistics.

Source	N species	P species represented	Meta-data Score (x/8)	N LH vars	P missing values (Key)	P missing values (All)
Amniotes	365	.68	8	23	.34	.54
AnAge	174	.33	3	19	.37	.46
Ernest	156	.29	8	9	.31	.32
Kappeler & Pereira	310	.58	5	13	.69	.76
PanTHERIA	376	.70	7	29	.63	.72

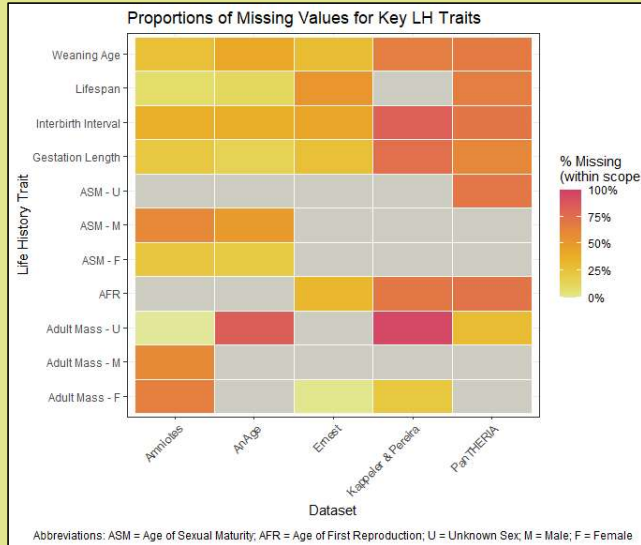


Figure 1. Heat map of missing values for key life history traits across all five databases. Low heat (yellow) signifies few missing values while high heat (red) signifies many missing values. Grey indicates that the trait was not recorded in a particular source.

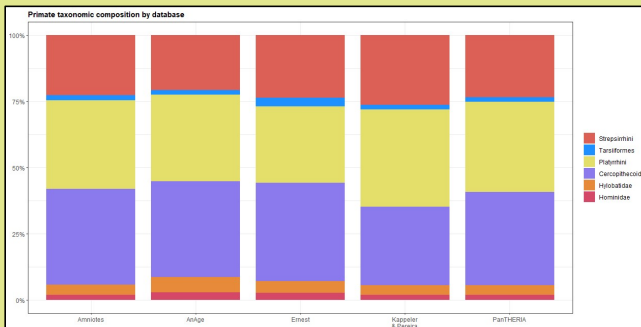


Figure 2. Relative proportions of different primate clades for each database.

Table 2. Example discrepancies in how key LH traits are recorded differently between databases. IBI = Inter-birth interval.

Variable	N unit types	Description	Database Comparison
Weaning Age	2	Mean duration from birth to solid food transition	Ernest [6] defines as eating solid food, PanTHERIA [9] as first solid food OR last nursing
IBI	4	Mean duration between successive births	Ernest [6] records litters/year, PanTHERIA [9] specifies between successful births
Gestation Length	2	Mean duration between conception and birth	Ernest [6] specifies live offspring, PanTHERIA [9] specifies "non-inactive" fetal growth

RESULTS

- No dataset represented **>70%** (376/535) of recognized primate species [12] (**Table 1**).
- At **least two** of our key life history traits were not recorded for each dataset. (**Figure 1**)
- Between datasets, mean proportions of missing values for key traits ranged from **30.7 to 68.8%**. (**Figure 1**)
- Of the traits recorded in multiple datasets, **AFR was the least complete among pooled data** with a proportion of missing values equal to .653.
- Despite dramatic differences in the number of species recorded, all datasets maintain relatively equal proportions of cladistic representation. (**Figure 2**)
- Metadata content varied from complete and detailed to largely unavailable. For those with available metadata, quality and detail also varied. (**Table 2**)

DISCUSSION

- Existing datasets lack consistency in which traits and species are included and how traits are defined and measured.
- Ex: Amniotes [11] and AnAge [7, 8] divide ASM by sex class, but not PanTHERIA [9]. Ernest [6] and Kappeler & Pereira [12] record AFR instead.
- Ex: The Amniotes [11] & PanTHERIA [9] definitions of IBI specify live offspring, while Kappeler & Pereira [12] doesn't specify; AnAge [7, 8] lacks any description. Ernest [6] does not record data for IBI.
- When collating many datasets, researchers should be aware of redundant data points and inconsistent recording methods.
- When compiling life history data, researchers should maintain thorough and accessible metadata to aid researchers using their dataset.
- We must establish protocols for collecting life history data and a standardized repository filled with reliable and accurate data.